MAPÚA UNIVERSITY

Laboratory Report 3

Object-Oriented Design and Implementation

Names:	Molar, Jabez C. Oabel, Edmarc Justin C.	**Date:**	September 6, 2026
Program:	CPE		

A. Learning Outcomes:

At the end of this laboratory, students should be able to:

- Apply Object-Oriented Design principles.
- Design classes and their relationships.
- Implement encapsulation, inheritance, abstraction, and polymorphism.
- Integrate multiple modules into a complete software system.
- Collaborate using GitHub Version Control.

B. Scenario:

A food delivery startup wants to develop a system that allows customers to browse menu items, place orders, process payments, and manage deliveries. The development team consists of four programmers, each responsible for a specific software module.

The goal is to design and implement the system using Object-Oriented Design principles and integrate all modules into a working application.

C. Project Requirements:

The system should be capable of:

1. Managing customers.
2. Managing menu items.
3. Creating food orders.
4. Processing payments.
5. Tracking deliveries.
6. Viewing completed transactions.

Please fill in the name of the PIC of each module:

1. Customer Management Module
2. Menu Management Module
3. Order Processing Module
4. Payment and Delivery Module

(Please include the following: **1.** Screenshots of the code **2.** Screenshots of the output) **Duplicate if necessary**

Github Repository Link: https://github.com/Jaabezt/CPE106L4-Food-Delivery-System.git

Laboratory Report 3

Object-Oriented Design and Implementation

Screenshots of Code (Name of Members) – Molar, Oabel

```python
class Customer:
    def __init__(self, customer_id, name, address, phone):
        self.__customer_id = customer_id
        self.__name = name
        self.__address = address
        self.__phone = phone

    def get_customer_id(self):
        return self.__customer_id

    def get_name(self):
        return self.__name

    def get_address(self):
        return self.__address

    def get_phone(self):
        return self.__phone

    def update_information(self, name, address, phone):
        self.__name = name
        self.__address = address
        self.__phone = phone

    def display_customer(self):
        print(f"Customer ID: {self.__customer_id}")
        print(f"Name: {self.__name}")
        print(f"Address: {self.__address}")
        print(f"Phone: {self.__phone}")
```

Laboratory Report 3

Object-Oriented Design and Implementation

```python
import json
import os

from customer import Customer
from order import Order
from payment import CashPayment, CardPayment
from delivery import Delivery
from transaction import Transaction


DATA_FILE = "data.json"


def save_data(customers, transactions):
    """
    Saves customers, orders, payments, deliveries,
    and completed transactions to a JSON file.
    """

    data = {
        "customers": {},
        "transactions": {}
    }

    # ========================
    # SAVE CUSTOMERS
    # ========================

    for customer_id, customer in customers.items():

        data["customers"][customer_id] = {
            "name": customer.get_name(),
            "address": customer.get_address(),
            "phone": customer.get_phone()
        }

    # ========================
    # SAVE TRANSACTIONS / ORDERS
    # ========================

    for order_id, record in transactions.items():

        order = record["order"]
        payment = record["payment"]
        delivery = record["delivery"]
        transaction = record["transaction"]

        order_data = {
            "order_id": order.get_order_id(),
            "customer_id": order.get_customer().get_customer_id(),
            "status": order.get_status(),
            "items": []
        }

        # Save order items
        for order_item in order.get_items():

            menu_item = order_item.get_menu_item()
```

```python
class Delivery:
    """
    Handles delivery tracking for an order.
    """
    def __init__(self, delivery_id, order_id, address):

        self.__delivery_id = delivery_id
        self.__order_id = order_id
        self.__address = address
        self.__status = "Preparing"



    def get_delivery_id(self):

        return self.__delivery_id



    def get_order_id(self):

        return self.__order_id



    def get_address(self):

        return self.__address



    def get_status(self):

        return self.__status



    def update_status(self, status):

        self.__status = status



    def display_delivery(self):

        print("\n========== DELIVERY ==========")

        print(f"Delivery ID: {self.__delivery_id}")
        print(f"Order ID: {self.__order_id}")
        print(f"Address: {self.__address}")
        print(f"Status: {self.__status}")

        print("==============================\n")
```

Laboratory Report 3

Object-Oriented Design and Implementation

menu.py × payment.py × menu_list.py × main.py × delivery.py × id_generator.py

menu_list.py > create_cafe_menu

```python
from menu import Menu, MenuItem

def create_cafe_menu():

    menu = Menu()


    # ========================
    # COFFEE
    # ========================

    coffee_items = [

        ("COF001", "Americano", 120.00, "Coffee"),
        ("COF002", "Cafe Latte", 150.00, "Coffee"),
        ("COF003", "Cappuccino", 160.00, "Coffee"),
        ("COF004", "Caramel Macchiato", 180.00, "Coffee"),
        ("COF005", "Spanish Latte", 170.00, "Coffee"),
        ("COF006", "Mocha Latte", 175.00, "Coffee"),
        ("COF007", "Vanilla Latte", 170.00, "Coffee")

    ]



    # ========================
    # NON COFFEE
    # ========================

    non_coffee_items = [

        ("DRK001", "Chocolate Frappe", 190.00, "Non-Coffee"),
        ("DRK002", "Matcha Latte", 180.00, "Non-Coffee"),
        ("DRK003", "Strawberry Milk", 160.00, "Non-Coffee"),
        ("DRK004", "Cookies and Cream Frappe", 200.00, "Non-Coffee"),
        ("DRK005", "Iced Lemon Tea", 130.00, "Non-Coffee"),
        ("DRK006", "Blueberry Yakult", 150.00, "Non-Coffee")

    ]



    # ========================
    # MAIN MEALS
    # ========================

    main_meals = [

        ("ML001", "Creamy Carbonara", 220.00, "Main Meal"),
        ("ML002", "Chicken Alfredo Pasta", 240.00, "Main Meal"),
        ("ML003", "Chicken Rice Bowl", 190.00, "Main Meal"),
        ("ML004", "Beef Teriyaki Bowl", 250.00, "Main Meal"),
        ("ML005", "Garlic Chicken Rice", 200.00, "Main Meal"),
        ("ML006", "Seafood Pasta", 260.00, "Main Meal")

    ]

```

Object-Oriented Design and Implementation

Terminal Help CPE106L4-Food-Delivery-System

menu.py × payment.py × menu_list.py × main.py × delivery.py × id_generator.py ×

menu.py > Menu > display_menu

```python
class MenuItem:
    """
    Represents a single menu item.
    """

    def __init__(self, item_id, name, price, category):

        # Encapsulation
        self.__item_id = item_id
        self.__name = name
        self.__price = float(price)
        self.__category = category

    # Getters

    def get_item_id(self):
        return self.__item_id

    def get_name(self):
        return self.__name

    def get_price(self):
        return self.__price

    def get_category(self):
        return self.__category

    # Update menu item

    def update_item(self, name, price, category):
        self.__name = name
        self.__price = float(price)
        self.__category = category

    # Display single item

    def display_item(self):
        print(
            f"{self.__item_id:<8}"
            f"{self.__name:<30}"
            f"₱{self.__price:>7.2f}"
        )


class Menu:
    """
    Handles multiple MenuItem objects.
    """

    def __init__(self):

        # Private list of MenuItem objects
        self.__items = []

    # Add item

    def add_item(self, item):
        self.__items.append(item)
```

Object-Oriented Design and Implementation

menu.py | order.py | payment.py | menu_list.py | main.py | delivery.py

order.py > ...

```python
from menu import MenuItem
from customer import Customer

class OrderItem:
    """
    Represents a menu item and its quantity in an order.
    """

    def __init__(self, menu_item, quantity):
        self.__menu_item = menu_item
        self.__quantity = int(quantity)

    def get_menu_item(self):
        return self.__menu_item

    def get_quantity(self):
        return self.__quantity

    def get_subtotal(self):
        return self.__menu_item.get_price() * self.__quantity


class Order:
    """
    Represents a customer's food order.
    """

    def __init__(self, order_id, customer):
        self.__order_id = order_id
        self.__customer = customer
        self.__items = []
        self.__status = "Pending"

    def get_order_id(self):
        return self.__order_id

    def get_customer(self):
        return self.__customer

    def get_items(self):
        return self.__items

    def get_status(self):
        return self.__status

    def add_item(self, menu_item, quantity):
        order_item = OrderItem(menu_item, quantity)
        self.__items.append(order_item)

    def remove_item(self, item_id):
        for order_item in self.__items:
            menu_item = order_item.get_menu_item()

            if menu_item.get_item_id() == item_id:
                self.__items.remove(order_item)
                return True

        return False
```

Object-Oriented Design and Implementation

payment.py > CardPayment > process_payment

```python
from abc import ABC, abstractmethod

class Payment(ABC):
    """
    Abstract payment class.
    """

    def __init__(self, payment_id, amount):

        self.__payment_id = payment_id
        self.__amount = float(amount)


    def get_payment_id(self):

        return self.__payment_id


    def get_amount(self):

        return self.__amount


    @abstractmethod
    def process_payment(self):

        pass



class CashPayment(Payment):
    """
    Cash payment implementation.
    """

    def __init__(self, payment_id, amount):

        super().__init__(payment_id, amount)



    def process_payment(self):

        return (
            f"Cash payment successful.\n"
            f"Payment ID: {self.get_payment_id()}\n"
            f"Amount: ₱{self.get_amount():.2f}"
        )



class CardPayment(Payment):
    """
    Card payment implementation.
    """

    def __init__(self, payment_id, amount):

```

```python
class Transaction:
    """
    Represents a completed food delivery transaction.
    """
    def __init__(self, transaction_id, order, payment, delivery):
        self.__transaction_id = transaction_id
        self.__order = order
        self.__payment = payment
        self.__delivery = delivery

    def get_transaction_id(self):
        return self.__transaction_id

    def display_transaction(self):
        print("\n========== TRANSACTION ==========")
        print(f"Transaction ID: {self.__transaction_id}")

        print("\nOrder:")
        print(f"Order ID: {self.__order.get_order_id()}")
        print(f"Customer: {self.__order.get_customer().get_name()}")
        print(f"Order Status: {self.__order.get_status()}")
        print(f"Order Total: ₱{self.__order.calculate_total():.2f}")

        print("\nPayment:")
        print(f"Payment ID: {self.__payment.get_payment_id()}")
        print(f"Payment Amount: ₱{self.__payment.get_amount():.2f}")

        print("\nDelivery:")
        print(f"Delivery ID: {self.__delivery.get_delivery_id()}")
        print(f"Delivery Status: {self.__delivery.get_status()}")

        print("=================================\n")
```

Screenshots of Output

```
========================================
          CAFE FOOD DELIVERIES
========================================
1. Customer Management
2. Menu Management
3. Create Food Order
4. Process Payment
5. Track Delivery
6. View Completed Transactions
7. Exit
========================================
Enter your choice: 1

========= CUSTOMER MANAGEMENT =========
1. Add Customer
2. View Customers
3. Update Customer
4. Back
Enter your choice: 1
Enter Name: Jabez Molar
Enter Address: QC
Enter Phone: 09738743646

Customer added successfully!
Customer ID: CUST001

========= CUSTOMER MANAGEMENT =========
1. Add Customer
2. View Customers
3. Update Customer
4. Back
Enter your choice: 4


========================================
          CAFE FOOD DELIVERIES
========================================
1. Customer Management
2. Menu Management
3. Create Food Order
4. Process Payment
5. Track Delivery
6. View Completed Transactions
7. Exit
========================================
Enter your choice: 2

========= MENU MANAGEMENT =========
1. Display Menu
2. Add Menu Item
3. Remove Menu Item
4. Back
Enter your choice: 1

========= CAFE MENU =========
```

```
========= CAFE MENU =========

--- COFFEE ---
COF001  Americano                     ₱ 120.00
COF002  Cafe Latte                    ₱ 150.00
COF003  Cappuccino                    ₱ 160.00
COF004  Caramel Macchiato             ₱ 180.00
COF005  Spanish Latte                 ₱ 170.00
COF006  Mocha Latte                   ₱ 175.00
COF007  Vanilla Latte                 ₱ 170.00

--- NON-COFFEE ---
DRK001  Chocolate Frappe              ₱ 190.00
DRK002  Matcha Latte                  ₱ 180.00
DRK003  Strawberry Milk               ₱ 160.00
DRK004  Cookies and Cream Frappe      ₱ 200.00
DRK005  Iced Lemon Tea                ₱ 130.00
DRK006  Blueberry Yakult              ₱ 150.00

--- MAIN MEAL ---
ML001   Creamy Carbonara              ₱ 220.00
ML002   Chicken Alfredo Pasta         ₱ 240.00
ML003   Chicken Rice Bowl             ₱ 190.00
ML004   Beef Teriyaki Bowl            ₱ 250.00
ML005   Garlic Chicken Rice           ₱ 200.00
ML006   Seafood Pasta                 ₱ 260.00

--- SNACK ---
SNK001  Clubhouse Sandwich            ₱ 210.00
SNK002  Tuna Sandwich                 ₱ 180.00
SNK003  French Toast                  ₱ 150.00
SNK004  Ham and Cheese Sandwich       ₱ 190.00
SNK005  Chicken Wrap                  ₱ 200.00

--- SIDE ---
SID001  French Fries                  ₱  90.00
SID002  Potato Wedges                 ₱ 140.00
SID003  Mozzarella Sticks             ₱ 160.00
SID004  Onion Rings                   ₱ 120.00
SID005  Nachos                        ₱ 170.00

===============================


========= MENU MANAGEMENT =========
1. Display Menu
2. Add Menu Item
3. Remove Menu Item
4. Back
Enter your choice: 2
Enter Item Name: Truffle Pasta
Enter Price: 550

Select Category:
1. Coffee
2. Non-Coffee
```

Object-Oriented Design and Implementation

```
========== MENU MANAGEMENT ==========
1. Display Menu
2. Add Menu Item
3. Remove Menu Item
4. Back
Enter your choice: 2
Enter Item Name: Truffle Pasta
Enter Price: 550

Select Category:
1. Coffee
2. Non-Coffee
3. Main Meal
4. Snack
5. Side
Enter your choice: 3

Menu item added successfully!
Item ID: ML007
Category: Main Meal

========== MENU MANAGEMENT ==========
1. Display Menu
2. Add Menu Item
3. Remove Menu Item
4. Back
Enter your choice: 1

========== CAFE MENU ==========

--- COFFEE ---
COF001  Americano                       ₱ 120.00
COF002  Cafe Latte                      ₱ 150.00
COF003  Cappuccino                      ₱ 160.00
COF004  Caramel Macchiato               ₱ 180.00
COF005  Spanish Latte                   ₱ 170.00
COF006  Mocha Latte                     ₱ 175.00
COF007  Vanilla Latte                   ₱ 170.00

--- NON-COFFEE ---
DRK001  Chocolate Frappe                ₱ 190.00
DRK002  Matcha Latte                    ₱ 180.00
DRK003  Strawberry Milk                 ₱ 160.00
DRK004  Cookies and Cream Frappe        ₱ 200.00
DRK005  Iced Lemon Tea                  ₱ 130.00
DRK006  Blueberry Yakult                ₱ 150.00

--- MAIN MEAL ---
ML001   Creamy Carbonara                ₱ 220.00
ML002   Chicken Alfredo Pasta           ₱ 240.00
ML003   Chicken Rice Bowl               ₱ 190.00
ML004   Beef Teriyaki Bowl              ₱ 250.00
ML005   Garlic Chicken Rice             ₱ 200.00
ML006   Seafood Pasta                   ₱ 260.00
ML007   Truffle Pasta                   ₱ 550.00

--- SNACK ---
SNK001  Clubhouse Sandwich              ₱ 210.00
```

```
--- SNACK ---
SNK001  Clubhouse Sandwich              ₱ 210.00
SNK002  Tuna Sandwich                   ₱ 180.00
SNK003  French Toast                    ₱ 150.00
SNK004  Ham and Cheese Sandwich         ₱ 190.00
SNK005  Chicken Wrap                    ₱ 200.00

--- SIDE ---
SID001  French Fries                    ₱  90.00
SID002  Potato Wedges                   ₱ 140.00
SID003  Mozzarella Sticks               ₱ 160.00
SID004  Onion Rings                     ₱ 120.00
SID005  Nachos                          ₱ 170.00

===============================


========== MENU MANAGEMENT ==========
1. Display Menu
2. Add Menu Item
3. Remove Menu Item
4. Back
Enter your choice: 4


========================================
          CAFE FOOD DELIVERIES
========================================
1. Customer Management
2. Menu Management
3. Create Food Order
4. Process Payment
5. Track Delivery
6. View Completed Transactions
7. Exit
========================================
Enter your choice: 3

========== CREATE FOOD ORDER ==========
Enter Customer ID: CUST001

========== CAFE MENU ==========

--- COFFEE ---
COF001  Americano                       ₱ 120.00
COF002  Cafe Latte                      ₱ 150.00
COF003  Cappuccino                      ₱ 160.00
COF004  Caramel Macchiato               ₱ 180.00
COF005  Spanish Latte                   ₱ 170.00
COF006  Mocha Latte                     ₱ 175.00
COF007  Vanilla Latte                   ₱ 170.00

--- NON-COFFEE ---
DRK001  Chocolate Frappe                ₱ 190.00
DRK002  Matcha Latte                    ₱ 180.00
DRK003  Strawberry Milk                 ₱ 160.00
DRK004  Cookies and Cream Frappe        ₱ 200.00
DRK005  Iced Lemon Tea                  ₱ 130.00
DRK006  Blueberry Yakult                ₱ 150.00
```

Object-Oriented Design and Implementation

```
PS C:\Users\oabel\OneDrive\Documents\CPE106L-4\CPE106L4-F
DRK004  Cookies and Cream Frappe      ₱ 200.00
DRK005  Iced Lemon Tea                ₱ 130.00
DRK006  Blueberry Yakult              ₱ 150.00

--- MAIN MEAL ---
ML001   Creamy Carbonara              ₱ 220.00
ML002   Chicken Alfredo Pasta         ₱ 240.00
ML003   Chicken Rice Bowl             ₱ 190.00
ML004   Beef Teriyaki Bowl            ₱ 250.00
ML005   Garlic Chicken Rice           ₱ 200.00
ML006   Seafood Pasta                 ₱ 260.00
ML007   Truffle Pasta                 ₱ 550.00

--- SNACK ---
SNK001  Clubhouse Sandwich            ₱ 210.00
SNK002  Tuna Sandwich                 ₱ 180.00
SNK003  French Toast                  ₱ 150.00
SNK004  Ham and Cheese Sandwich       ₱ 190.00
SNK005  Chicken Wrap                  ₱ 200.00

--- SIDE ---
SID001  French Fries                  ₱  90.00
SID002  Potato Wedges                 ₱ 140.00
SID003  Mozzarella Sticks             ₱ 160.00
SID004  Onion Rings                   ₱ 120.00
SID005  Nachos                        ₱ 170.00

===============================

Enter Item ID to add (or type DONE to finish): SID001
Enter Quantity: 5

French Fries added to order.

========== CAFE MENU ==========

--- COFFEE ---
COF001  Americano                     ₱ 120.00
COF002  Cafe Latte                    ₱ 150.00
COF003  Cappuccino                    ₱ 160.00
COF004  Caramel Macchiato             ₱ 180.00
COF005  Spanish Latte                 ₱ 170.00
COF006  Mocha Latte                   ₱ 175.00
COF007  Vanilla Latte                 ₱ 170.00

--- NON-COFFEE ---
DRK001  Chocolate Frappe              ₱ 190.00
DRK002  Matcha Latte                  ₱ 180.00
DRK003  Strawberry Milk               ₱ 160.00
DRK004  Cookies and Cream Frappe      ₱ 200.00
DRK005  Iced Lemon Tea                ₱ 130.00
DRK006  Blueberry Yakult              ₱ 150.00

--- MAIN MEAL ---
ML001   Creamy Carbonara              ₱ 220.00
ML002   Chicken Alfredo Pasta         ₱ 240.00
ML003   Chicken Rice Bowl             ₱ 190.00
ML004   Beef Teriyaki Bowl            ₱ 250.00
ML005   Garlic Chicken Rice           ₱ 200.00
ML006   Seafood Pasta                 ₱ 260.00
```

```
ML003   Chicken Rice Bowl             ₱ 190.00
ML004   Beef Teriyaki Bowl            ₱ 250.00
ML005   Garlic Chicken Rice           ₱ 200.00
ML006   Seafood Pasta                 ₱ 260.00
ML007   Truffle Pasta                 ₱ 550.00

--- SNACK ---
SNK001  Clubhouse Sandwich            ₱ 210.00
SNK002  Tuna Sandwich                 ₱ 180.00
SNK003  French Toast                  ₱ 150.00
SNK004  Ham and Cheese Sandwich       ₱ 190.00
SNK005  Chicken Wrap                  ₱ 200.00

--- SIDE ---
SID001  French Fries                  ₱  90.00
SID002  Potato Wedges                 ₱ 140.00
SID003  Mozzarella Sticks             ₱ 160.00
SID004  Onion Rings                   ₱ 120.00
SID005  Nachos                        ₱ 170.00

===============================

Enter Item ID to add (or type DONE to finish): DONE

========== ORDER ==========
Order ID: ORD001
Customer: Jabez Molar
Status: Pending

Items:
French Fries x 5 = ₱450.00

Total: ₱450.00
===========================

Confirm this order? (Y/N): Y

Order confirmed!
Order ID: ORD001


========================================
          CAFE FOOD DELIVERIES
========================================
1. Customer Management
2. Menu Management
3. Create Food Order
4. Process Payment
5. Track Delivery
6. View Completed Transactions
7. Exit
========================================
Enter your choice: 4

========== PROCESS PAYMENT ==========
Enter Order ID: ORD001

Amount to pay: ₱450.00

1. Cash
```

Object-Oriented Design and Implementation

```
PS C:\Users\Dabel\OneDrive\Documents\CPE106L-4\CPE
Amount to pay: ₱450.00

1. Cash
2. Card
Select payment method: 1

Cash payment successful.
Payment ID: PAY001
Amount: ₱450.00

Payment processed successfully!
Payment ID: PAY001


========================================
          CAFE FOOD DELIVERIES
========================================
1. Customer Management
2. Menu Management
3. Create Food Order
4. Process Payment
5. Track Delivery
6. View Completed Transactions
7. Exit
========================================
Enter your choice: 5

========= TRACK DELIVERY =========
Enter Order ID: ORD001

Delivery created!
Delivery ID: DEL001

========= DELIVERY =========
Delivery ID: DEL001
Order ID: ORD001
Address: QC
Status: Preparing
==============================

1. Out for Delivery
2. Delivered
3. Back
Enter your choice: 1

Delivery status updated.

========= DELIVERY =========
Delivery ID: DEL001
Order ID: ORD001
Address: QC
Status: Out for Delivery
==============================

1. Out for Delivery
2. Delivered
3. Back
Enter your choice: 2

Order delivered!
```

```
Order delivered!


========================================
          CAFE FOOD DELIVERIES
========================================
1. Customer Management
2. Menu Management
3. Create Food Order
4. Process Payment
5. Track Delivery
6. View Completed Transactions
7. Exit
========================================
Enter your choice: 6

========= COMPLETED TRANSACTIONS =========

========= TRANSACTION =========
Transaction ID: TXN001

Order:
Order ID: ORD001
Customer: Jabez Molar
Order Status: Delivered
Order Total: ₱450.00

Payment:
Payment ID: PAY001
Payment Amount: ₱450.00

Delivery:
Delivery ID: DEL001
Delivery Status: Delivered
=================================

============================================


========================================
          CAFE FOOD DELIVERIES
========================================
1. Customer Management
2. Menu Management
3. Create Food Order
4. Process Payment
5. Track Delivery
6. View Completed Transactions
7. Exit
========================================
Enter your choice: 7

Thank you for using the Cafe Food Deliveries!
```

MAPÚA UNIVERSITY

Laboratory Report 3

Object-Oriented Design and Implementation

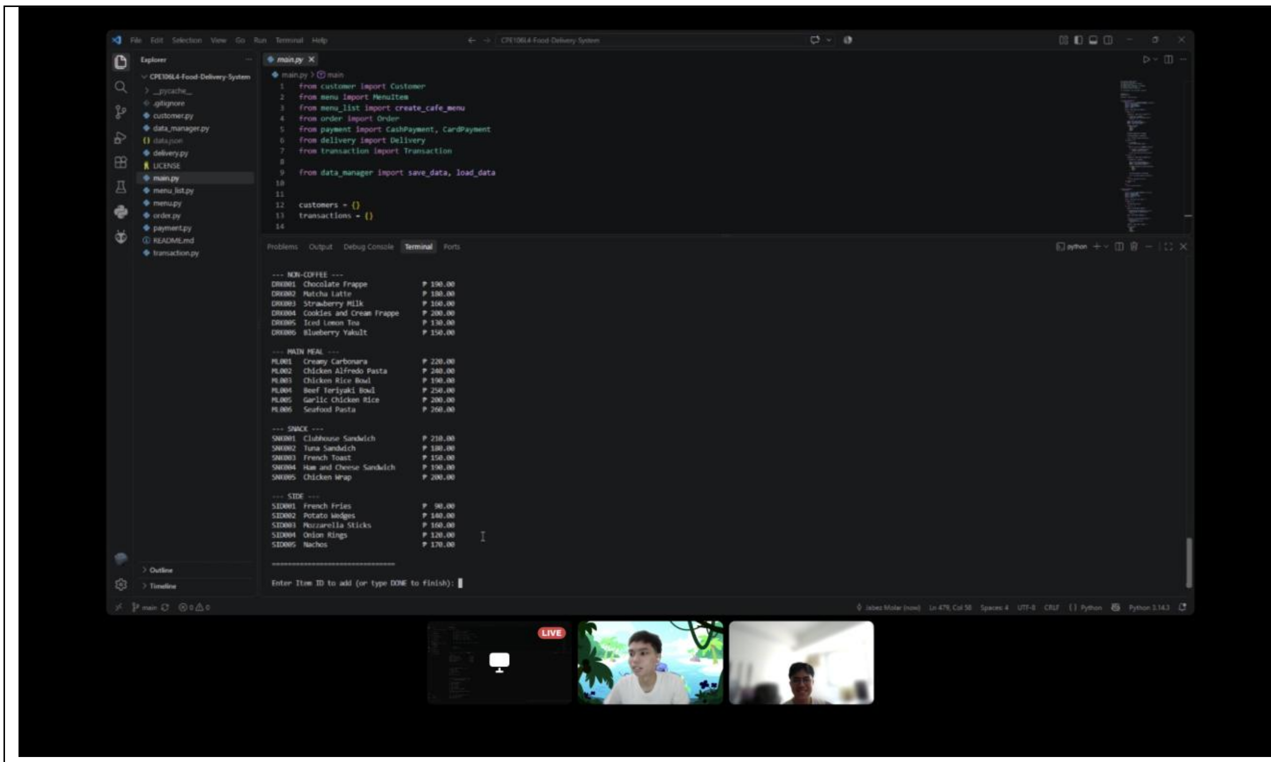
D. Findings, Observations, and Comments:

Guide Questions:

1. What are your Group's realizations in this laboratory?
2. How did you apply your learnings from CPE104 Data Structures and Algorithms in this activity?

Our group realized that proper data structures, system organization, and clear class responsibilities are essential in developing reliable and maintainable software. We applied Object-Oriented Design by designing related classes and implementing encapsulation, inheritance, abstraction, and polymorphism while integrating multiple modules into one complete system. From our learnings in CPE104 Data Structures and Algorithms, we applied appropriate data structures and algorithms to efficiently store, locate, access, and process information, including the use of unique identifiers for better data management. During testing, we identified bugs and areas for improvement, such as finding better ways to implement features like adding menu items. We also collaborated using GitHub Version Control by working on two separate branches to divide our tasks, manage our individual parts, and coordinate our work smoothly before integrating the system. Through this laboratory, we strengthened our understanding of OOP, data structures, system development, and collaborative programming.

PICTURE/PROOF OF DOING THE ACTIVITY



Object-Oriented Design and Implementation

E. Score Sheet

Criteria	Poor (25%)	Fair (50%)	Good (75%)	Excellent (100%)	Score
I. Completeness, Documentation, and Organization of Report	The report was incomplete, messy, and had many unanswered questions.	The report was complete, messy, and had many unanswered questions.	The report was complete, neat, and had 1 or 2 unanswered questions.	The report was complete, neat, and all the questions had been answered.	
II. Coding Design and Patterns	The program had inappropriate elements and structures, incorrect indention, and poor program documentation.	The program had corrected indention, but inappropriate elements and structures and poor program documentation.	The program had corrected indention, adequate program documentation, but inappropriate elements and structures.	The program had appropriate elements and structures, correct indention, and excellent program documentation.	
III. Findings, Observations, and Comments	The findings. observations, and comments were not based on the gathered data and results. All thoughts were inconsistent and unclear.	The findings. observations, and comments based on the gathered data and results but were not elaborated. Not all thoughts were consistent and clear.	The findings. observations, and comments were based on the gathered data and results and were mostly elaborated. Most of the thoughts were consistent and clear.	The findings. observations, and comments were based on the gathered data and results and were completely elaborated. All the thoughts were consistent and clear.	
IV. Grammar and Composition	The words used were inappropriate, wrong grammar usage, and poor sentence construction was observed.	The words used were appropriate; however, bad grammar and poor sentence construction were observed.	The words used were appropriate, had proper grammar usage, but poor sentence construction was observed.	The words used were appropriate, had proper grammar usage, and excellent sentence construction.	
SCORE:					

Laboratory Report 3

Object-Oriented Design and Implementation

CHECKED			
Rating		**Signature**	
		Date	